do with the output. This is because the agent was given a high-level overview of what it accomplished. Remember, though, that plan execution is done at the local level, and only context, plan, and goal were sent to the LLM.

This means that plan execution can be completed by any process, not necessarily by the agent. Executing a plan outside the LLM reduces the tokens and tool use the agent needs to perform. This also means that an LLM doesn't need to support tools usage to use a planner.

Internally, when a planner is enabled within Nexus, the agent engine tool is bypassed. Instead, the planner completes the action execution, and the agent is only aware of the actions through the passing of the output context. This can be good for models that support tool use but can't plan. However, a planner may limit functionality for models that support both tool use and planning, such as Claude.

In general, you'll want to understand the capabilities of the LLM you're using. If you're unsure of those details, then a little trial and error can also work. Ask the agent to complete a multistep goal with and without planning enabled, and then see the results.

Planning allows agents to complete multiple sequential tasks to achieve more complex goals. The problem with external or prompt planning is that it bypasses the feedback iteration loop, which can help correct problems quickly. Because of this, OpenAI and others are now directly integrating reasoning and planning at the LLM level, as we'll see in the next section.

11.4 Reviewing a stepwise planner: OpenAI Strawberry

The release of the o1-preview model, code named Strawberry, introduced a dramatic shift in the type of LLMs becoming available for agentic systems. Strawberry was not only proclaimed to be more efficient at math, science, and general calculation tasks but also able to engage in reasoning, planning, evaluation, and feedback directly in the LLM.